

Data Analyst Job Postings

Analysis of Experience and Programming Languages Requirements

Introduction

This report analyzes a sample of 400 U.S. data-analyst job postings, each scraped from online listings and enriched using a large language model to extract two structured fields from the free-text job description: the minimum years of experience required and which data programming languages (R, Python, both, or neither) are mentioned. The goal is to explore how the advertised median salary estimate relates to experience requirements and to the programming-language skills a posting calls for.

Years of Experience Analysis

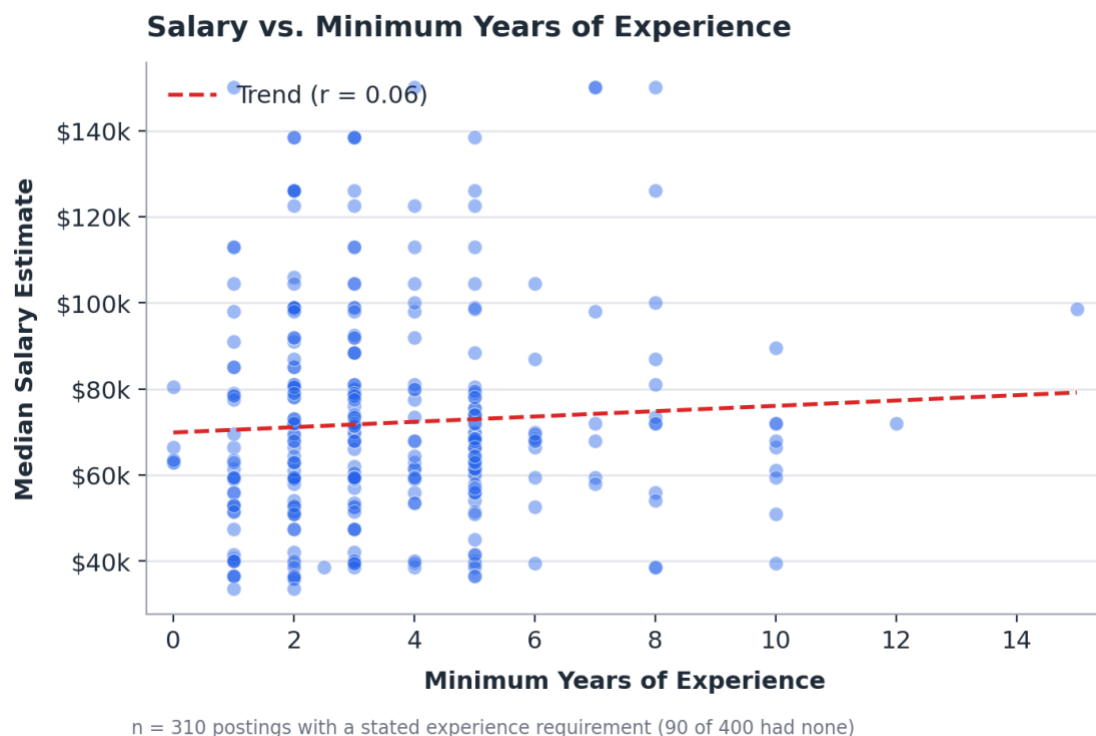


Figure 1. Minimum years of experience vs. median salary estimate ($n = 310$ postings that stated an experience requirement).

The scatter plot shows almost no relationship between the minimum years of experience a posting requires and its median salary estimate. The Pearson correlation is only $r = 0.06$, and the fitted trend line is nearly flat, implying a barely perceptible increase of roughly \$600 in median salary per additional year of required experience. Postings clustered at the common thresholds of 2, 3, and 5 years span almost the full salary range, from the low \$30,000s to roughly \$140,000, so experience alone does little to explain how these jobs are paid.

Two limitations temper this reading. First, the sample is small and, importantly, 90 of the 400 postings (about 22%) did not state any experience requirement and were excluded from the plot; if those missing values are not random, the visible pattern could be biased. Second, “minimum years” is a coarse, self-reported figure that ignores seniority, location, industry, and company size, all of which plausibly drive salary more strongly than a stated experience floor.

Programming Language Analysis

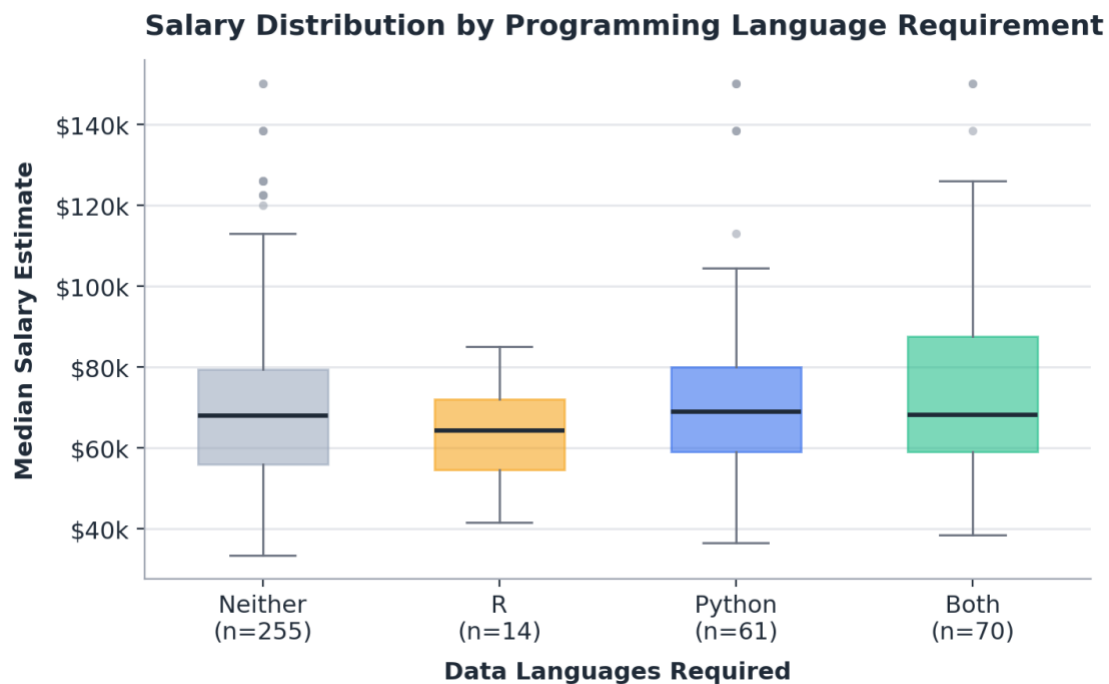


Figure 2. Median salary estimate by required data languages (group sizes shown beneath each box).

Grouping postings by the data languages they require, the median salaries are strikingly similar across categories, sitting between roughly \$64,000 and \$69,000 for all four groups. Postings requiring Python (\$69,000) or both R and Python (\$68,250) have median salaries marginally above those mentioning neither language (\$68,000) or R alone (\$64,250). The spread, however, differs: the “Both” and “Neither” groups reach the highest salaries near \$140,000–\$150,000, while postings requiring only R have both the lowest median and the narrowest range.

These differences should be read cautiously because the groups are very unevenly sized. Only 14 postings require R alone, so its box is built on too few data points to be reliable, whereas “Neither” accounts for 255 of the 400 postings. The heavy concentration of postings that mention no programming language at all suggests many data-analyst roles in this sample emphasize tools such as Excel, SQL, or BI platforms rather than R or Python, and any apparent salary premium for programming skills is small relative to the overall variation within each group.

Reflection

One part of this assignment that stood out to me was how important it was to separate data cleaning from data analysis. At first I assumed the main work would be making the graphs and interpreting them, but I realized the analysis only made sense once the extracted fields had been cleaned and standardized. The "Minimum Years of Experience" column had to be handled carefully, since some postings did not give a clear number, and those values could not be plotted as if they were real requirements. The "Data Languages" column also had to be standardized into the four categories — R, Python, Both, and Neither — so the charts were built on consistent groups rather than messy text. Keeping the LLM extraction, the cleaning, and the analysis as separate stages made the project feel more organized, because each step had its own purpose: the model pulled information out of the descriptions, cleaning made that information usable, and the analysis explained the patterns. It also made me trust the results more, since I knew the charts reflected consistent categories and real numbers.

The results themselves surprised me. I expected postings that required more years of experience to clearly pay more, but there was almost no relationship between minimum experience and median salary — the correlation was only about 0.06. I also found it interesting that "Neither" was by far the largest group, meaning many data analyst postings did not require R or Python at all. This made me realize how much these roles vary, and that salary likely depends on factors the data didn't fully capture, such as location, industry, company size, or other tools like SQL, Excel, and BI software. Overall, I learned that an LLM can be genuinely useful for pulling structured information out of messy job descriptions, but its outputs still need to be checked, cleaned, and organized before any meaningful analysis can happen.